

Deepanshu Pathak

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github.com/deepanshu-iitm | [Portfolio](#)

EDUCATION

Indian Institute of Technology, Madras
Bachelor of Science, Data Science and Applications

Chennai, Tamil Nadu
2021 – Present

PROJECTS

The Code Archaeologist – AI-Powered Codebase Analysis Assistant

GraphRAG + Neo4j + LLM (Gemini) — Python, FastAPI, tree-sitter

- Designed and implemented an **AI-powered multi-language code analysis system** using **tree-sitter** for AST parsing across 7+ programming languages (Python, JavaScript, TypeScript, Java, C++, Go, Rust).
- Built a **Neo4j-based knowledge graph** to represent code structures (files, classes, functions, variables, imports) and their relationships, enabling deep semantic code queries.
- Developed a **GraphRAG query engine** integrating **Google Gemini** for natural language → Cypher query translation, automated query planning, and answer synthesis from graph data.
- Implemented a **FastAPI-based web interface** and **CLI tools** for repository ingestion, querying, and graph statistics, with support for real-time query suggestions.

AI-Powered Legal Document Review Agent

FastAPI, React.js, ChromaDB, Google Gemini, SentenceTransformers, PyMuPDF

- Developed an AI-driven legal document assistant capable of analyzing contracts, agreements, and compliance documents with high accuracy.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB and SentenceTransformers for context-aware querying.
- Integrated Google Gemini API with custom-built tools for automated extraction of key dates, parties, clauses, and legal terms.
- Engineered multi-format ingestion supporting PDF and DOCX with efficient text parsing and preprocessing.
- Built a scalable FastAPI backend and a responsive React.js frontend for a seamless user experience.

AutoResearcher – AI-Powered Research Automation Platform

FastAPI, React.js, ChromaDB, SentenceTransformers, Python

- Engineered a scalable, asynchronous backend using **FastAPI** to manage concurrent data collection, AI processing, and report generation for a seamless research workflow.
- Developed a Retrieval-Augmented Generation (RAG)-ready pipeline using **ChromaDB** and **SentenceTransformers** to create a high-performance semantic search engine for academic literature.
- Designed a modular, multi-source data ingestion system to asynchronously collect and process documents from **arXiv**, **Semantic Scholar**, and **Wikipedia**, featuring a smart deduplication algorithm to ensure data quality.
- Implemented an automated report generation engine to synthesize findings into structured Markdown and professional **Word documents (.doc)**.

ProjectTracker | *React, Flask, PostgreSQL, GitHub API, OpenAI API, GitHub Actions*

- Developed a web application to monitor student progress in software development projects
- Integrated GitHub for real-time tracking of coding activity and commit histories
- Implemented Generative AI to analyze project documents and generate insights, predicting potential project delay
- Designed a centralized dashboard for instructors to manage projects, milestones, and team progress

HACKATHONS

Agentic Ethereum Hackathon 2024 – Participant

Built *The Oasis*, a Web3-based AI filmmaking platform that leverages generative AI for automated script-to-scene production. Integrated blockchain-based ownership for creative assets, combining AI pipelines with Ethereum smart contracts.

ISRO Bhartiya Antariksha Hackathon 2024 – Participant

Designed a geospatial web portal to estimate rooftop solar energy potential using deep learning for building footprint extraction. Leveraged satellite imagery, solar datasets, and interactive visualization tools to deliver building-level solar energy assessments.

TECHNICAL SKILLS

Languages: Java, Python, Javascript, Typescript, C/C++, SQL

Machine Learning : Scikit-Learn, XGBoost, Matplotlib, Seaborn, Pandas, Numpy, SentenceTransformers, Retrieval-Augmented Generation (RAG), GraphRAG, LLM, NLP, Langchain

Web Development : Flask, React.js, Next.js, Vue.js, TailwindCSS, HTML/CSS

Tools and Frameworks : Git, GitHub, JIRA, Anaconda, VS Code, Jupyter Notebooks, Docker

Database Systems : PostgreSQL, SQLite, MongoDB